

TECHNICAL UNIVERSITY OF CRETE

School of Electrical & Computer Engineering

Investigating the Applicability of Federated Dynamic Averaging in Federated Low-Rank Adaptation Architectures

Diploma Thesis Presentation

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Background Knowledge : Distributed Learning, FedAvg & Client Drift

- **Federated Learning (FL)**: Privacy-preserving collaborative optimization of foundation models without centralizing client data.
- **Federated Averaging (FedAvg)**:

$$w_{t+1} = w_t + \frac{1}{|S_t|} \sum_{k \in S_t} \Delta_t^{(k)}$$

Clients execute τ local SGD steps on private data $\mathcal{D}_k$.

- **Statistical Heterogeneity (Non-IID)**: Client distributions differ in label, quantity, or features. Modeled via **Dirichlet distribution** $p_k \sim \mathrm{Dir}(\alpha)$.
- **Client Drift**: Divergent local minima pull updates away from the true global optimum.

Dirichlet Heterogeneity (α)

- $\alpha \to 0$: Extreme label skew (clients hold single-class data).
- $\alpha \to \infty$: Uniform IID distribution.
- Key benchmark lever for evaluating drift robustness.

The Dual Communication Bottleneck

- **Spatial**: Model parameter volume (billions of weights).
- **Temporal**: Rigid synchronization cadence ($\tau = E$).

Background Knowledge : Low-Rank Adaptation (LoRA)

- Pre-trained weights $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ are kept **frozen**.
- Updates factorized into low-rank matrices:

$$\Delta W = \frac{\alpha_{\text{lora}}}{r} BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \; B \in \mathbb{R}^{d_{\text{out}} \times r}$$

  where rank $r \ll \min(d_{\text{in}}, d_{\text{out}})$.
- Forward pass: $h = W_0 x + \frac{\alpha_{\text{lora}}}{r} BAx$.
- Trainable parameters reduced by $> 99\%$ with zero inference latency overhead.
- **Structural & Initialization Asymmetry**:
  - **Down-Projection** $A \sim \mathcal{N}(0, \sigma^2)$: Compresses inputs to rank r.
  - **Up-Projection** $B = 0$: Expands to d_{out} (ensures $\Delta W = 0$ at $t = 0$).
  - **Asymmetric Gradients**: At onset, $\nabla_A \mathcal{L} \propto B^\top \nabla_h \mathcal{L} = 0$ while $\nabla_B \mathcal{L} \neq 0$, inducing divergent drift scales across A and B.

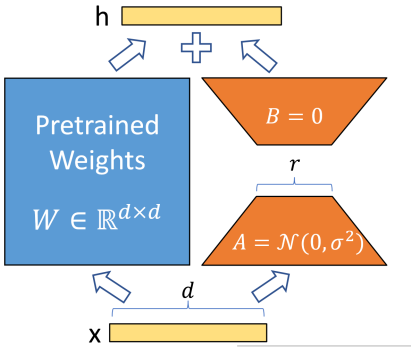


Figure: LoRA decomposition with frozen weights W_0 (Hu et al., 2022).

Background Knowledge : The LoRA Aggregation Inconsistency Problem

Mathematical Non-Commutativity (Bilinear Distortion)

Independent server-side averaging of client low-rank factors fails to reconstruct the composite weight update:

$$\frac{1}{N}\sum_{i=1}^{N}(B_i A_i) \neq \left(\frac{1}{N}\sum_{i=1}^{N} B_i\right)\left(\frac{1}{N}\sum_{i=1}^{N} A_i\right)$$

- Direct FedAvg on (B_i, A_i) induces cross-client geometric interference.
- Destabilizes convergence under non-IID data distributions (Yang et al., IJCAI 2025).
- **Prior Work Limitations**:
  - *Full-matrix expansion* ($\Delta W = BA$): Transmits full parameter matrices $\rightarrow$ destroys spatial savings.
  - *FFA-LoRA*: Permanently freezes down-projection A $\rightarrow$ sacrifices adapter expressivity.
  - *FLoRA*: Server-side SVD recalculation $\rightarrow$ high server compute overhead.

Alternating Coordinate Optimization Restores Aggregation Linearity

RoLoRA resolves the bilinear inconsistency by alternating the active trainable matrix while freezing the other globally across all clients:

- **Even Rounds (Train B, Freeze A)**: $\frac{1}{N}\sum_{i=1}^{N}(B_iA) = \left(\frac{1}{N}\sum_{i=1}^{N}B_i\right)A = \bar{B}A$
- **Odd Rounds (Train A, Freeze B)**: $\frac{1}{N}\sum_{i=1}^{N}(BA_i) = B\left(\frac{1}{N}\sum_{i=1}^{N}A_i\right) = B\bar{A}$
- **Core Limitation**: Enforces a rigid, static local epoch schedule (E), leaving temporal synchronization completely unoptimized.

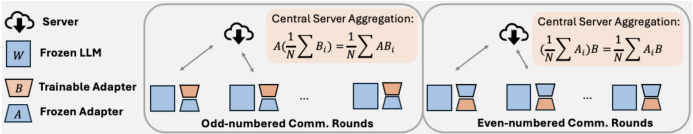


Figure: RoLoRA alternating framework across communication rounds (Chen et al. schematic uses row-vector AB notation; thesis follows standard column-vector BA convention).

Background Knowledge : FDA-Opt & Geometric Monitoring

Theologitis, Samoladas & Deligiannakis, 2025 – Dynamic Variance Halting

- **Geometric Monitoring Root**: Decomposes global threshold functions over distributed streams into local **safe zones** (Sharfman et al., 2006).
- **FDA-Opt Formulation** (Theologitis et al., 2025): Tracks client update dispersion via parameter variance:

$$\mathrm{Var}(W_{t,i}) = \frac{1}{|S_t|} \sum_{k \in S_t} \|\Delta_{t,i}^{(k)}\|_2^2 - \|g_{t,i}\|_2^2$$

  Low variance = aligned progress; high variance = client drift.
- **Event-Driven Halting**: Synchronize when $\mathrm{Var}(W) > \theta_t$.
- **Midpoint Threshold Scaling**: $\theta_{t+1} = \mathrm{var}_t \cdot \frac{\tau}{2s_t}$.

Research Gap in FDA-Opt

- FDA-Opt operated strictly on **full model parameters** (approximated via AMS sketches).
- Spatial compression (LoRA) and temporal adaptivity (FDA) had never been synthesized.
- Low-rank adapters exhibit high variance volatility requiring decoupled mechanics.

Background Knowledge : Synthesis of Foundational Precursors

Synthesizing RoLoRA (Spatial) and FDA-Opt (Temporal)

Dimension	**RoLoRA** (NeurIPS '25)	**FDA-Opt** (2025)	**FDA-RoLoRA** (**This Thesis**)
Optimization Focus	Spatial parameter volume	Temporal synchronization	**Joint Spatial + Temporal**
Parameter Scope	Low-rank adapters (A, B)	Full model weights	**Low-rank adapters (A, B)**
Aggregation	Alternating coordinates	Standard FedAvg / FedOpt	**Alternating coordinates**
Local Scheduling	Static fixed (E epochs)	Dynamic (Model variance)	**Dynamic (Subspace variance)**
Variance Tracking	None	Full-weight (AMS sketch)	Exact in simulation (monitoring cost not evaluated)
Threshold Logic	None	Single scalar threshold	**Decoupled Dual-Track** (θ_A, θ_B)

Complementary Synergy

FDA-RoLoRA solves the core limitations of both parent papers: RoLoRA provides the strict aggregation linearity needed for stable variance tracking, while FDA-Opt provides the event-driven cadence needed to eliminate static schedule waste.

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Methodology : Dynamic Averaging in Federated LoRA

Taxonomy of Integration Avenues

Different PEFT architectures have distinct mathematical structures and optimization geometries; dynamic averaging cannot use a single one-size-fits-all rule.

- **FDA + FedAvg-LoRA**: Dynamic averaging with simultaneous (A, B) training.
  - *Drawback*: Inherits bilinear aggregation error; non-linear distortion destabilizes variance monitoring.
- **FDA + FFA-LoRA**: Dynamic averaging with permanently frozen A *[Future Work]*.
- **FDA + SLoRA**: Dynamic thresholding over sparse adapter coordinates *[Future Work]*.
- **FDA-RoLoRA**: **This Thesis**
  - Synthesizes **RoLoRA** (Chen et al., 2025) and **FDA-Opt** (Theologitis et al., 2025).
  - Alternating coordinates provide strict linearity $\rightarrow$ perfect substrate for variance tracking.
  - Maintains independent persistent variance thresholds (θ_A, θ_B) for each subspace.

Methodology : FDA-RoLoRA Architecture

Decoupled Dual-Track Subspace Variance Mechanics

RoLoRA isolates the optimization trajectories of down-projection (A) and up-projection (B). Because A and B exhibit severe gradient scale disparity ($\theta_B \gg \theta_A$), FDA-RoLoRA maintains two persistent, decoupled threshold tracks (θ_A, θ_B).

- **Active Subspace Mapping**: Server maps active threshold based on designated matrix $\mathcal{M}^{(r)} \in \{B, A\}$ (B-first schedule):

$$\theta_{\text{current}}^{(r)} = \begin{cases} \theta_A & \text{if } \mathcal{M}^{(r)} = A \\ \theta_B & \text{if } \mathcal{M}^{(r)} = B \end{cases}$$

- **Direct Subspace Variance**: Evaluated exclusively on low-rank coordinate tensors:

$$\text{Var}_{\text{subspace}}(\Delta^{(e)}) = \frac{1}{|S_r|} \sum_{i \in S_r} \|\Delta_i^{(e)}\|_2^2 - \left\| \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e)} \right\|_2^2$$

- Eliminates memory and compute overhead of projecting drifts back to full weight space $\Delta W = \frac{\alpha}{r} BA$.

Methodology : Independent Halting & Local Training Bounds

Local Epoch Bounds ($[e_{\min}, e_{\max}]$)

Client training bounded in $[e_{\min}, e_{\max}]$ with operational floor $e_{\min}$ to guarantee baseline progress:

$$\text{Halt if } \text{Var}_{\text{subspace}}(\Delta^{(e)}) > \theta^{(r)}_{\text{current}} \ \vee \ e \geq e_{\max}$$

Evaluated at epoch boundaries $e \geq e_{\min}$.

Operational Progress Floor ($e_{\min} = 1$)

- Ensures clients execute at least one local epoch before variance checks begin.
- Both tracks initialize $\theta_A = \theta_B = 0$, calibrating baseline variance at $e^* = 1$ in rounds 0 and 1.

Subspace Monitoring

Calculated exclusively on the active low-rank coordinate matrix (A or B):

$$\text{Var}_{\text{subspace}}(\Delta^{(e)}) = \frac{1}{|S_r|} \sum_{i \in S_r} \|\Delta_i^{(e)} - \bar{\Delta}^{(e)}\|_2^2$$

- Evaluated over $< 0.5\%$ of backbone parameters (120 KB per active track vs. 500 MB full model).
- With sketches in a deployment, the full adapter uplink (120 KB) would be sent once per round at e^*. Sketches were not run in these experiments.

Simulation vs. Deployment

In simulation, exact low-rank variance is computed directly in memory. A real deployment must send monitoring data at every epoch check. This cost is not included in the reported MB.

Methodology : Coordinate-Wise FedAvg & Protocol Flow

Coordinate-Wise FedAvg Aggregation

- Client drift: $\Delta_i^{(e^*)} = W_{i,e^*}^{(r)} - W_{\text{act}}^{(r)}$
- Cohort average: $\bar{\Delta}^{(r)} = \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e^*)}$
- Linear FedAvg update ($\eta_s = 1.0$):

$$W_{\text{act}}^{(r+1)} = W_{\text{act}}^{(r)} + \bar{\Delta}^{(r)}$$

- **Linear Aggregation**: Standard FedAvg avoids momentum artifacts, isolating gains strictly to dynamic variance tracking.

Protocol Execution Lifecycle

1. **Toggle**: Server selects active track A or B.
2. **Broadcast**: Sends $W_{\text{act}}^{(r)}$ and $\theta_{\text{current}}^{(r)}$ to S_r.
3. **Train**: Clients run local SGD steps ($\eta_c = 0.1$).
4. **Check**: Evaluate subspace variance at $e \geq e_{\min}$ (exact and in memory in simulation; a deployment must send monitoring data at each check).
5. **Halt**: Return $\Delta_i^{(e^*)}$ when $\text{Var} > \theta$ or $e = e_{\max}$.
6. **Aggregate**: Server averages client drifts ($\bar{\Delta}^{(r)}$).
7. **Evolve**: Update active threshold $\theta_{\mathcal{M}^{(r)}}^{(\text{next})}$.
8. **Step**: Server applies FedAvg to active coordinate.

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Implementation : Modular 3-Tier Software Architecture

1. Server Orchestrator

- Global A, B matrices.
- Tracks (θ_A, θ_B) state.
- Coordinates RoLoRA toggle.
- Coordinate-wise FedAvg.

2. Client Nodes

- Private data partitions.
- Local SGD training loops.
- Subspace drift computation.
- Variance checks at $e \geq e_{\min}$.

3. Comm. Interface

- Low-rank serialization.
- PyTorch, HF & PEFT.
- Lightweight FDA-Opt core.

Key Engineering Decisions

- **FDA-Opt Base vs. FederatedScope-LLM**: The RoLoRA paper (Chen et al., NeurIPS 2025) used FederatedScope-LLM without a public release. We adopted and extended the open-source **FDA-Opt** repository (Theologitis et al., 2025), integrating HuggingFace PEFT and RoLoRA coordinate alternation into its modular PyTorch core.
- **Exact Variance in Simulation**: Variance is computed exactly in memory to evaluate the synchronization rule under ideal measurement. This does not remove the need for monitoring communication in a real deployment.

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Evaluation : Experimental Setup & Benchmark Protocol

Foundation Model Configuration, Low-Rank Parameterization & Benchmark Protocol

Algorithm | Hardware Details

Model & Low-Rank Parameterization

- **Model Backbone**: `roberta-base` (125M params, 12 layers, 768 hidden; selected for multi-seed sweeps under hardware budget).
- **Adapter Configuration**: Rank $r = 4$, scaling factor $\alpha_{\text{lora}} = 8$ on Query (W_q) and Value (W_v) projections across top blocks $\{7 \dots 11\}$ (MNLI, QNLI, QQP) and top 3 blocks $\{9 \dots 11\}$ (SST-2); **about 0.05% trainable parameter share**.
- **Bi-Directional Adapter Payload**: 240 KB/client/round (4.69 MB at $C = 20$, 11.72 MB at $C = 50$); 144 KB for SST-2. **Monitoring data is excluded**.
- **Optimization**: Client SGD ($\eta_c = 0.1$, batch $B \in \{32, 64\}$); Server FedAvg ($\eta_s = 1.0$).

Benchmark Regimes & Baselines

- **Tasks & Datasets**: GLUE benchmark suite (MNLI, QNLI, QQP, SST-2) capped at $\leq 25{,}000$ training samples; full validation sets.
- **Cohorts & Regimes**: $C \in \{20, 50\}$ clients (full participation); Homogeneous ($\alpha = 1000.0$) vs. Dirichlet Label Skew ($\alpha = 0.5$).
- **Baselines**: Fixed RoLoRA $E \in \{5, 10, 15, 20\}$ where run. Non-IID uses $E = 10$ in every configuration, plus $E = 20$ on MNLI and QNLI ($C = 20$).
- **FDA-RoLoRA Settings**: $e_{\min} = 1$, $e_{\max} \in \{20, 40\}$, $k \in \{0.5, 1.0\}$.
- **Execution Budget**: **173 completed runs** evaluated under uniform platform (**2 seeds per configuration**).

Evaluation : Results Summary

All GLUE Benchmarks on `roberta-base`

Full IID Table | Full Non-IID Table

Task & Cohort	Regime	Top FDA-RoLoRA ($e_{\min} = 1$)	FDA Peak (%)	FDA Final (%)	Top RoLoRA Baseline	RoLoRA Peak (%)	RoLoRA Final (%)
MNLI ($C = 20$)	IID ($\alpha = 1000$)	FDA ($e_{\max} = 20, k = 1.0$)	**76.86 ± 0.10**	**76.86 ± 0.10**	RoLoRA ($E = 10$)	76.60 ± 0.85	76.60 ± 0.85
	Non-IID ($\alpha = 0.5$)	FDA ($e_{\max} = 20, k = 1.0$)	**74.47 ± 0.11**	73.71 ± 0.77	RoLoRA ($E = 10$)	74.24 ± 0.12	**74.11 ± 0.31**
MNLI ($C = 50$)	IID ($\alpha = 1000$)	FDA ($e_{\max} = 20, k = 1.0$)	**69.85 ± 3.37**	**69.85 ± 3.37**	RoLoRA ($E = 10$)	65.85 ± 2.85	65.85 ± 2.85
	Non-IID ($\alpha = 0.5$)	FDA ($e_{\max} = 20, k = 0.5$)	55.56 ± 1.65	55.23 ± 2.11	RoLoRA ($E = 10$)	**56.55 ± 6.94**	**56.22 ± 7.40**
QNLI ($C = 20$)	IID ($\alpha = 1000$)	FDA ($e_{\max} = 20, k = 1.0$)	**83.14 ± 0.34**	67.63 ± 22.28	RoLoRA ($E = 10$)	81.83 ± 0.92	**75.03 ± 10.54**
	Non-IID ($\alpha = 0.5$)	FDA ($e_{\max} = 20, k = 1.0$)	**84.22 ± 0.41**	**81.88 ± 2.90**	RoLoRA ($E = 10$)	83.50 ± 1.67	81.10 ± 4.31
QNLI ($C = 50$)	IID ($\alpha = 1000$)	FDA ($e_{\max} = 20, k = 1.0$)	82.77 ± 2.14	82.47 ± 1.72	RoLoRA ($E = 20$) (1 seed)	**88.14 ± 0.00**	**87.94 ± 0.00**
	Non-IID ($\alpha = 0.5$)	FDA ($e_{\max} = 20, k = 0.5$)	**75.69 ± 1.66**	50.13 ± 0.94	RoLoRA ($E = 10$)	74.61 ± 10.15	**58.45 ± 12.71**
QQP ($C = 20$)	IID ($\alpha = 1000$)	FDA ($e_{\max} = 20, k = 1.0$)	**78.57 ± 4.23**	**76.34 ± 5.33**	RoLoRA ($E = 10$)	75.47 ± 1.19	75.47 ± 1.19
	Non-IID ($\alpha = 0.5$)	FDA ($e_{\max} = 20, k = 1.0$)	**77.88 ± 0.66**	64.52 ± 18.23	RoLoRA ($E = 10$)	77.69 ± 2.18	**76.76 ± 3.41**
QQP ($C = 50$)	IID ($\alpha = 1000$)	FDA ($e_{\max} = 20, k = 0.5$)	73.86 ± 0.39	72.08 ± 0.20	RoLoRA ($E = 10$)	**78.79 ± 1.38**	**75.43 ± 1.07**
	Non-IID ($\alpha = 0.5$)	FDA ($e_{\max} = 20, k = 1.0$)	74.55 ± 1.30	57.45 ± 11.29	RoLoRA ($E = 10$)	**74.71 ± 1.63**	**69.28 ± 8.14**
SST-2 ($C = 20$)	IID ($\alpha = 1000$)	FDA ($e_{\max} = 20, k = 1.0$)	92.14 ± 0.24	91.80 ± 0.24	RoLoRA ($E = 20$)	**92.55 ± 0.30**	**92.16 ± 0.13**
	Non-IID ($\alpha = 0.5$)	FDA ($e_{\max} = 20, k = 1.0$)	**89.91 ± 0.81**	69.72 ± 29.19	RoLoRA ($E = 10$)	89.62 ± 1.87	**89.62 ± 1.87**
SST-2 ($C = 50$)	IID ($\alpha = 1000$)	FDA ($e_{\max} = 20, k = 1.0$)	89.68 ± 1.14	89.68 ± 1.14	RoLoRA ($E = 20$)	**92.01 ± 0.57**	**91.90 ± 0.37**
	Non-IID ($\alpha = 0.5$)	FDA ($e_{\max} = 20, k = 1.0$)	**85.95 ± 5.60**	**81.36 ± 9.97**	RoLoRA ($E = 10$)	59.00 ± 11.11	53.33 ± 3.08

Table Note: QNLI $C = 20$ Non-IID: RoLoRA $E = 20$ final accuracy is 82.97 ± 0.50.

Key Empirical Takeaways

Peak vs. Final: Against each configuration's best static baseline, FDA-RoLoRA wins 10 of 16 on peak accuracy but 4 of 16 on final accuracy (Non-IID: 6 of 8 vs. 1 of 8).

MNLI Non-IID ($C = 20$): Reaches RoLoRA $E = 10$ final accuracy in 13–30% fewer rounds, while avoiding the $E = 20$ post-peak drop (73.31% → 68.87%). A byte saving is not established once monitoring is counted.

Plateau Escape on SST-2 Non-IID ($C = 50$): Static RoLoRA trapped near majority-class (59.00% peak, 53.33% final); FDA dynamically expands compute ($\bar{e} = 14.57$) to reach **85.95**% peak (+**26.95**% gain).

Evaluation : Homogeneous Consensus (IID, $\alpha = 1000.0$)

No Consistent Advantage: The Controller Undershoots Its Target

MNLI/QNLI Plots | QQP/SST-2 Plots

Optimization Dynamics & Controller Behavior

- **Consensus**: Updates align along shared trajectory ($\bar{\nu} \leq 1.58$).
- **Super-Linear Drift Surge**: In compact low-rank subspaces, coordinate variance grows super-linearly ($\mathrm{Var}_{\mathrm{subspace}}(t) \propto t^p$, $p > 1$).
- **Setpoint Undershoot**: Breaches linear predictor early $\rightarrow$ systematic undershoot ($\bar{e} \approx 5.8$–7.8 for $k = 0.5$; $\bar{e} \approx 9.1$–13.4 for $k = 1.0$).

Observations

- **QQP ($C = 50$)**: Stops early at rd 14 at a lower plateau (72.65% final, 73.63% peak) vs. RoLoRA $E = 10$ (75.43% final, 78.79% peak) $\rightarrow$ earlier plateau, not bandwidth saving at parity.
- **MNLI ($C = 50$)**: Deeper compute ($\bar{e} = 12.15$) breaks static $E = 10$ plateau (69.85% vs. 65.85%).
- **Takeaway**: FDA-RoLoRA shows no consistent advantage over fixed-epoch RoLoRA under IID data.

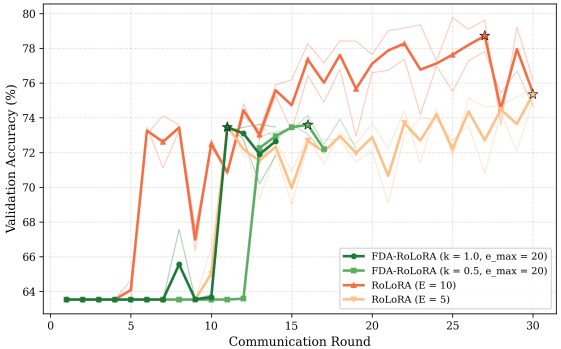


Figure: Validation accuracy vs. communication rounds on QQP ($C = 50$, IID).

Evaluation : Statistical Heterogeneity (Non-IID, $\alpha = 0.5$)

Where FDA-RoLoRA Helps: Plateau Escape and Fewer MNLI Rounds

MNLI & Epoch Plots

Controller Behaviour Under Label Skew

- Gradient dispersion across skewed client classes linearizes drift accumulation across local epochs.
- Controller tracks nominal setpoint closely ($\mathcal{S} \approx 0.91$–1.11, $\bar{e} \approx 8.7$–10.9 for $k = 0.5$ and $\bar{e} \approx 13.0$–17.5 for $k = 1.0$).
- Expands local training when variance is low and synchronizes earlier when it rises. This did not prevent late accuracy drops.

Empirical Highlights

- **SST-2 ($C = 50$)**: RoLoRA ($E = 10$) trapped near majority class (59.00% peak, 53.33% final); FDA expands compute ($\bar{e} = 14.57$) to reach **85.95**% peak (+**26.95**% gain!).
- **MNLI ($C = 20$)**: Reaches RoLoRA $E = 10$ final accuracy in 13–30% fewer rounds and avoids the $E = 20$ post-peak drop.
- **QNLI**: Reaches 84.22% peak at $C = 20$. At $C = 50$, it reaches 75.69% peak with $k = 0.5$ (RoLoRA: 74.61%), but both FDA-RoLoRA variants end near chance (49.51% and 50.13%).

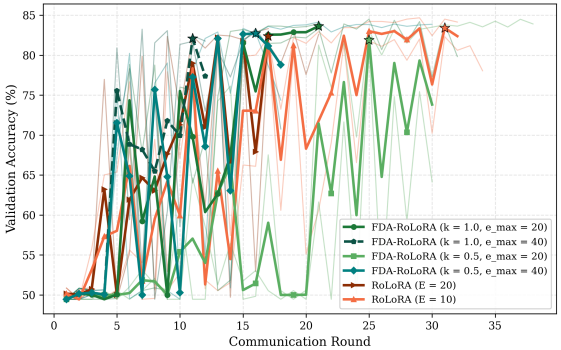


Figure: Validation accuracy vs. communication rounds on QNLI under Dirichlet skew ($\alpha = 0.5$, $C = 20$).

Evaluation : Stability: Peak Accuracy Overstates the Benefit

Disparity Between Peak and Final Convergence

- 13 of 38 Non-IID FDA-RoLoRA runs ended at least 10 points below their peak, compared with 3 of 20 RoLoRA runs.
- Under IID data: 2 of 32 FDA-RoLoRA runs and 2 of 46 RoLoRA runs.
- SST-2 ($C = 20, k = 1.0$): one seed fell from 89.3% to 49.1%.
- QNLI ($C = 50$): both FDA-RoLoRA variants ended near chance.
- QQP ($C = 20$): final accuracy 64.52% for FDA-RoLoRA vs. 76.76% for RoLoRA $E = 10$.
- **Exception**: on SST-2 ($C = 50$), FDA-RoLoRA ended above RoLoRA in both seeds.

Evaluation : Rounds and Data to Reach Accuracy Gates

Non-IID ($\alpha = 0.5$), FDA-RoLoRA with $k = 1.0$

Gate	RoLoRA $E = 10$	RoLoRA $E = 20$	FDA adapter MB	FDA with sketches
MNLI ($C = 20$) $\geq 65\%$	131.2 MB (28 rd)	70.3 MB (15 rd)	84.4 MB (18 rd)	116.3 MB
MNLI ($C = 20$) $\geq 70\%$	164.1 MB (35 rd)	86.8 MB (18.5 rd)	107.8 MB (23 rd)	150.4 MB
MNLI ($C = 20$) final accuracy (73.17%/74.25%)	227.3 MB (48.5 rd)	Fell to 68.87%	178.1 MB (38 rd)	254.2 MB
SST-2 ($C = 20$) $\geq 85\%$	64.7 MB (23 rd)	Not run	23.9 MB (8.5 rd)	35.2 MB

Table Note: Adapter MB excludes monitoring. Sketch figures add 6 KB per client per epoch and assume the same training trajectory.

Faster Than E = 10

FDA-RoLoRA reaches the MNLI gates in fewer rounds than $E = 10$, and reaches final accuracy in 13–30% fewer rounds. $E = 20$ reaches the 65% and 70% gates sooner, then falls to 68.87%. On SST-2 ($C = 20$), it reaches 85% in 8.5 vs. 23 rounds, but one seed later fell to 49%.

SST-2 (C = 50) Plateau Escape

RoLoRA $E = 10$ stays near the majority-class baseline (final 51.15% and 55.50%). FDA-RoLoRA ends at 88.42% and 74.31% within the same 20 rounds. One of two seeds reached 85%.

Evaluation : Monitoring Cost: What the MB Figures Exclude

Monitoring	**Added cost on top of adapter MB**	**MNLI Non-IID total (RoLoRA $E = 10$: 227.3 MB)**
Not counted (thesis)	None	178.1 MB (-22%)
6 KB sketch per epoch	$+15\%$ to $+59\%$	254.2 MB ($+12\%$)
Exact upload per epoch	$3.5\times$ to $9.2\times$	1,645 MB ($7.2\times$)

Impact of Monitoring Overhead

- Correction: the 2.5–5.8% overhead in the thesis assumes one check per round, but FDA-RoLoRA checks every epoch.
- Break-even with 6 KB sketches: about 20% fewer rounds at $\bar{e} = 10$, and about 30% at $\bar{e} = 17.5$, at equal accuracy.
- Sketch figures assume the same training trajectory. No real sketch run was performed.

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Conclusion : Synthesis of Scientific Contributions

Core Theoretical and Empirical Insights from FDA-RoLoRA

- **Preservation of Aggregation Linearity**:
  - Updating both adapter matrices simultaneously incurs bilinear aggregation error ($\frac{1}{N}\sum B_i A_i \neq \bar{B}\bar{A}$), distorting client updates and misguiding variance tracking.
  - Alternating coordinate optimization isolates one factor (A or B) per round, maintaining strictly linear updates and enabling stable geometric drift measurement.

- **Scale-Decoupled Geometric Monitoring**:
  - Tracking variance directly on low-rank coordinate tensors avoids the memory cost of full-model estimation. Its communication cost was not measured.
  - Decoupled factor thresholds (θ_A, θ_B) resolve coordinate scale disparity while preserving symmetrical local epoch progression ($\bar{e}_A \approx \bar{e}_B$).

- **Empirical Foundation Model Benchmarks (`roberta-base`, 173 Runs)**:
  - **Non-IID Label Skew ($\alpha = 0.5$)**: Controller tracks setpoint closely ($\mathcal{S} \approx 0.91$–1.11); reaches target accuracy in 13–30% fewer rounds on MNLI ($C = 20$) and escapes the majority plateau on SST-2 ($C = 50$, $+26.95\%$).
  - **IID Consensus ($\alpha = 1000$)**: Controller undershoots the target epoch setpoint ($\mathcal{S} \approx 0.64$–0.79); demonstrates no consistent communication advantage over fixed-epoch RoLoRA.
  - **Communication**: Round savings were observed, but a byte saving against RoLoRA is not established.
  - **Stability**: Several non-IID runs lost accuracy late, so peak accuracy overstates the benefit.

Conclusion : Limitations & Future Research Horizons

Practical & Experimental Limitations

- **RoLoRA Reimplementation**: Original codebase unavailable; reimplemented inside FDA-Opt framework.
- **Hyperparameter Search**: RoLoRA paper omitted key LRs; required extensive empirical tuning.
- **Compute Constraints**: Evaluated on `roberta-base` (125M); multi-billion LLMs remain future targets.
- **Simulated FL**: Multi-process on 2×A10 GPUs; real WAN latency and dropouts unmodeled.
- **Monitoring Cost**: Per-epoch monitoring communication was not measured, and real sketches were not run.

Future Research Horizons

- **Larger Foundation Models**: Scale to autoregressive LLMs (LLaMA, Mistral) on instruction-tuning.
- **Broader PEFT Integration**: Synthesize FDA with SLoRA, FFA-LoRA, and dynamic rank (AdaLoRA).
- **Partial Participation**: Evaluate under subsampled client cohorts and alternative factor alternation orders.
- **Ecosystem Integration**: Upstream native FDA-RoLoRA support into FederatedScope-LLM.
- **Monitoring Evaluation**: Run FDA-RoLoRA with real AMS sketches and LinearFDA, and measure total communication.

Key References

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Thank You!

Questions & Discussion

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Open-Source Repository: `https://github.com/Asterinos1/FDA-RoLoRA`

Backup Slides (Q&A Quick Jump):

Algorithm & Early Stop | Hardware & Testbed | Threshold Evolution | Factor Asymmetry

MNLI Non-IID Plots | Full Non-IID Table | Full IID Table | IID Plots (1) | IID Plots (2)

Backup : FDA-RoLoRA Algorithm & Early-Stopping Harness

Complete Pseudocode (Algorithm 3) & Simulation Control Parameters

Back to Setup Q&A

Algorithm 3: FDA-RoLoRA Execution Loop

1. **Init**: $W = W_0 + \frac{\alpha}{r} B_0 A_0$; $\theta_A = \theta_B = 0$; decoupled server optimizers for A and B.
2. **For** round $r = 0, 1, 2, \ldots$:
   - Toggle active matrix $\mathcal{M}^{(r)} \in \{B, A\}$ (B on even rounds, A on odd).
   - Load active threshold $\theta^{(r)}_{\text{current}} \leftarrow \theta_{\mathcal{M}^{(r)}}$ and broadcast $W^{(r)}_{\text{act}}$.
   - **For** local epoch $e = 1 \ldots e_{\max}$:
     - Each client $i \in S_r$ steps $W^{(r)}_{i,e} \leftarrow W^{(r)}_{i,e-1} - \eta_c \nabla \mathcal{L}_i$ and computes $\Delta^{(e)}_i = W^{(r)}_{i,e} - W^{(r)}_{\text{act}}$.
     - **If** $e \geq e_{\min}$ and $(\text{Var}_{\text{subspace}}(\Delta^{(e)}) > \theta^{(r)}_{\text{current}}$ or $e = e_{\max})$: set $e^* \leftarrow e$ and **break**.
   - Server computes $\nabla^{(r)}_{\text{pseudo}} = -\frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e^*)}$ and updates $W^{(r+1)}_{\text{act}}$.
   - Update active threshold $\theta_{\mathcal{M}^{(r)}} \leftarrow \left(\frac{k \cdot e_{\max}}{e^*}\right) \text{Var}_{\text{subspace}}(\Delta^{(e^*)})$ (dormant track frozen).

Note: Variance is exact in simulation; monitoring communication is not counted.

Accuracy-Gated Early Stopping (Sec. 4.3)

- **Sliding-Window Smoothing (W)**: Trailing moving average over $W = 3$ (IID) or $W = 5$ (Non-IID) rounds to filter stochastic non-IID oscillations.
- **Accuracy Activation Gate (A_{gate})**: Disarmed during initial coordinate calibration until validation accuracy crosses task milestone.
- **Patience (P) & Min Gain ($\Delta_{\min}$)**: Terminates if smoothed accuracy fails to improve by $\Delta_{\min} = 0.5\%$ over $P \in \{3, 5\}$ rounds.
- **Mandatory Round Guard ($R_{\min}$)**: Enforces minimum alternating cycles before any stop trigger.

Backup : Testbed Implementation & Hardware Environment

Physical Infrastructure, Execution Orchestration & Client Constraints

◂ Back to Setup ◂ Q&A

Polytechnix Server Testbed

- **Server**: SoftNet Lab Polytechnix GPU server.
- **GPUs**: 2× NVIDIA Ampere A10 (24 GB VRAM each).
- **CPUs**: 2× Intel Xeon Silver 4310 (24c/48t, 2.10 GHz).
- **RAM**: 256 GB DDR4 system memory.
- **Comparability**: All 173 runs evaluated exclusively on this uniform platform to ensure perfect comparability.

Simulation Pipeline Orchestration

- **Alternation**: Track B on even rounds ($t = 0, 2, \dots$), Track A on odd rounds ($t = 1, 3, \dots$).
- **Client Simulation**: Sequential in-silico GPU client simulation (constraining VRAM to one active client) paired with 6 concurrent jobs (3/GPU) across 2× A10 GPUs.
- **Batch Sizes**: $B = 32$ (MNLI, QNLI, QQP), $B = 64$ (SST-2, following Chen et al., 2025).
- **Client Partitions**: Dirichlet-sampled private data splits with frozen backbone weights.

Generalized Proportional Setpoint Control Law

Active threshold scales proportionally to realized epoch e^* relative to target setpoint $k \cdot e_{\max}$ ($k \in (0, 1]$), while dormant track remains frozen:

$$\theta^{(\text{next})}_{\mathcal{M}^{(r)}} = \left(\frac{k \cdot e_{\max}}{e^*} \right) \text{Var}_{\text{subspace}}(\Delta^{(e^*)}) \quad \text{where } k = 0.5 \implies \frac{e_{\max}}{2} \text{ (Midpoint)}, \; k = 1.0 \implies e_{\max} \text{ (Ceiling)}$$

- **Self-Stabilizing Feedback**: Early halt ($e^* < k e_{\max}$) expands θ next cycle; late halt ($e^* > k e_{\max}$) tightens θ.

Evolutionary Design Decisions (Explored & Discarded in Pilot Study)

- **Simultaneous FDA + FedLoRA**: Suffered bilinear product error ($\frac{1}{N} \sum B_i A_i \neq \bar{B} \bar{A}$) $\rightarrow$ premature accuracy plateaus.
- **Dense Reconstruction (FDA-LoRA-SVD)**: Server-side truncated SVD on $\sum B_i A_i$ restored accuracy, but $\mathcal{O}(d^2)$ communication and $\mathcal{O}(d^2 r)$ SVD latency proved prohibitive.
- **Single Shared Threshold (θ)**: Cross-matrix scale contamination ($\theta_B \gg \theta_A$) caused erratic halting across tracks.
- **Temporal Smoothing (EMA / SMA)**: Introduced phase lag across alternating subspaces; memoryless feedback proved superior.

Backup : Factor Asymmetry & Decoupled Tracking

Local Epoch Symmetry ($\bar{e}_A \approx \bar{e}_B$) & Threshold Decoupling

◂ Q&A

Scale Disparity, Not Deeper Training

- **Down-Projection** A: Gaussian initialization.
- **Up-Projection** B: Zero initialization ($\Delta W = 0$).
- Initialization induces severe numerical scale disparity ($\theta_B \gg \theta_A$).
- Decoupled thresholds (θ_A, θ_B) are essential to prevent a shared threshold from choking A or over-training B.

Empirical Symmetry

Across all Non-IID benchmarks (Table 5.2), decoupled thresholds yield remarkably **symmetrical** epoch progression across factors ($\bar{e}_A \approx \bar{e}_B$).

Non-IID Setting ($\alpha = 0.5$)	$\mathbf{e}_{\max}$	**k**	$\mathbf{e}^*$	~~$\mathbf{e}$~~	$\mathcal{S}$	~~$\mathbf{e_A}$~~ /~~$\mathbf{e_B}$~~
MNLI ($C = 20$)	20	0.50	10.0	10.07	1.01	9.96 / 10.18
MNLI ($C = 20$)	20	1.00	20.0	17.49	0.87	17.19 / 17.76
MNLI ($C = 20$)	40	0.50	20.0	19.06	0.95	18.82 / 19.29
QNLI ($C = 20$)	20	1.00	20.0	16.73	0.84	16.63 / 16.81
QNLI ($C = 20$)	40	1.00	40.0	29.83	0.75	30.14 / 29.53
SST-2 ($C = 50$)	20	1.00	20.0	14.57	0.73	14.60 / 14.55

Aggregate Regime Tracking Fidelity

Across steady-state horizons, Non-IID achieves high setpoint tracking ($\mathcal{S} = 1.11 \pm 0.08$ at $k = 0.5$; $\mathcal{S} = 0.91 \pm 0.05$ at $k = 1.0$), whereas IID undershoots ($\mathcal{S} = 0.79$ at $k = 0.5$; $\mathcal{S} = 0.64$ at $k = 1.0$).

Backup : MNLI Non-IID Plots & Epoch Dynamics

Dirichlet Label Skew ($\alpha = 0.5, C = 20$): 6-Panel Plots & Setpoint Tracking

◂ Back to Non-IID ◂ Q&A

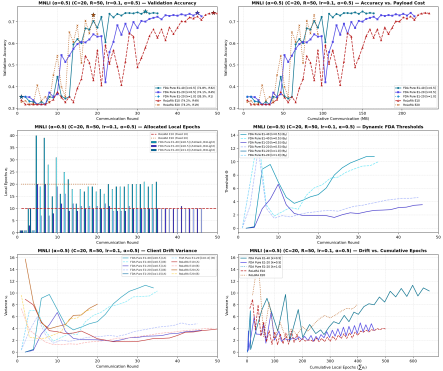
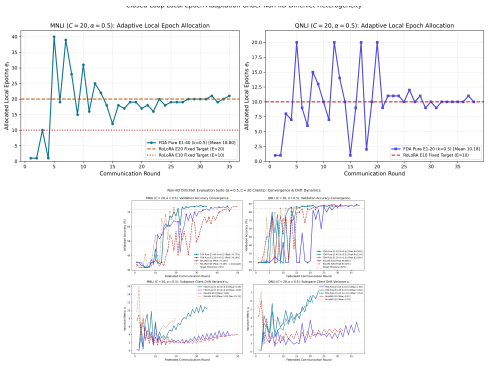


Fig. 5.9: MNLI Non-IID ($C = 20$): FDA-RoLoRA reaches 74.47% peak and reaches final accuracy in 13–30% fewer rounds than RoLoRA $E = 10$, while RoLoRA $E = 20$ collapses (73.31% $\rightarrow$ 68.87%).



Figs. 5.11 & 5.12: Top: Realized local epochs per round hovering around $k \cdot e_{\max}$ on MNLI ($e_{\max} = 40, k = 0.5$) and QNLI ($e_{\max} = 20, k = 0.5$). Bottom: Accuracy and coordinate drift variance σ_t^2.

Backup : Complete Non-IID Benchmark Table (Table 5.2)

All Evaluated Non-IID Configurations ($\alpha = 0.5$) Across GLUE Benchmarks

◂ Back to Results Summary ◂ Q&A

Task	Cohort	Methodology	Final Acc. (%)	Peak Acc. (%)	Rounds (R)	Total MB	Mean $\bar{e}$	Mean $\bar{e}_A$	Mean $\bar{e}_B$
MNLI	$C = 20$ ($R_{\max} = 50$)	FDA-RoLoRA E1TO40 ($k = 0.5$)	73.70 ± 0.34	74.12 ± 0.93	34.5 ± 0.7	161.7 MB	19.06	18.82	19.29
		FDA-RoLoRA E1TO20 ($k = 0.5$)	72.48 ± 0.14	73.25 ± 1.19	41.0 ± 7.1	192.2 MB	10.07	9.96	10.18
		FDA-RoLoRA E1TO20 ($k = 1.0$)	73.71 ± 0.77	$\mathbf{74.47 \pm 0.11}$	38.0 ± 4.2	178.1 MB	17.49	17.19	17.76
		RoLoRA E20	68.87 ± 6.46	73.31 ± 0.18	$\mathbf{24.0 \pm 5.7}$	**112.5** MB	20.00	20.00	20.00
		RoLoRA E10	$\mathbf{74.11 \pm 0.31}$	74.24 ± 0.12	50.0	234.4 MB	10.00	10.00	10.00
	$C = 50$ ($R_{\max} = 30$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	51.46 ± 8.46	54.95 ± 3.53	$\mathbf{22.5 \pm 2.1}$	**263.7** MB	15.63	15.03	16.20
		FDA-RoLoRA E1TO20 ($k = 0.5$)	55.23 ± 2.11	55.56 ± 1.65	30.0	351.6 MB	10.13	9.47	10.80
		RoLoRA E10	$\mathbf{56.22 \pm 7.40}$	$\mathbf{56.55 \pm 6.94}$	30.0	351.6 MB	10.00	10.00	10.00
QNLI	$C = 20$ ($R_{\max} = 50$)	FDA-RoLoRA E1TO40 ($k = 1.0$)	76.27 ± 7.78	83.20 ± 0.18	$\mathbf{15.0 \pm 4.2}$	**70.3** MB	29.83	30.14	29.53
		FDA-RoLoRA E1TO40 ($k = 0.5$)	79.12 ± 6.72	83.48 ± 0.94	24.0 ± 8.5	112.5 MB	20.36	21.24	19.47
		FDA-RoLoRA E1TO20 ($k = 1.0$)	81.88 ± 2.90	$\mathbf{84.22 \pm 0.41}$	26.5 ± 7.8	124.2 MB	16.73	16.63	16.81
		FDA-RoLoRA E1TO20 ($k = 0.5$)	74.03 ± 13.98	82.64 ± 2.63	34.0 ± 5.7	159.4 MB	10.54	10.81	10.27
		RoLoRA E20	$\mathbf{82.97 \pm 0.50}$	83.22 ± 0.14	19.0 ± 2.8	89.1 MB	20.00	20.00	20.00
		RoLoRA E10	81.10 ± 4.31	83.50 ± 1.67	33.0 ± 1.4	154.7 MB	10.00	10.00	10.00
	$C = 50$ ($R_{\max} = 30$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	49.51 ± 0.08	73.52 ± 4.31	**13.0**	**152.3** MB	13.04	13.17	12.93
		FDA-RoLoRA E1TO20 ($k = 0.5$)	50.13 ± 0.94	$\mathbf{75.69 \pm 1.66}$	30.0	351.6 MB	10.48	9.83	11.13
		RoLoRA E10	$\mathbf{58.45 \pm 12.71}$	74.61 ± 10.15	30.0	351.6 MB	10.00	10.00	10.00
QQP	$C = 20$ ($R_{\max} = 20$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	64.52 ± 18.23	$\mathbf{77.88 \pm 0.66}$	14.0 ± 2.8	65.6 MB	15.70	15.77	15.62
		FDA-RoLoRA E1TO20 ($k = 0.5$)	48.32 ± 12.63	73.58 ± 1.56	$\mathbf{12.0 \pm 4.2}$	**56.2** MB	10.87	11.79	10.10
		RoLoRA E10	$\mathbf{76.76 \pm 3.41}$	77.69 ± 2.18	16.5 ± 0.7	77.3 MB	10.00	10.00	10.00
	$C = 50$ ($R_{\max} = 24$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	57.45 ± 11.29	74.55 ± 1.30	$\mathbf{15.5 \pm 3.5}$	**181.6** MB	14.26	13.83	14.63
		FDA-RoLoRA E1TO20 ($k = 0.5$)	53.46 ± 19.74	72.71 ± 3.23	$\mathbf{15.5 \pm 4.9}$	**181.6** MB	9.18	8.89	9.48
		RoLoRA E10	$\mathbf{69.28 \pm 8.14}$	$\mathbf{74.71 \pm 1.63}$	22.5 ± 2.1	263.7 MB	10.00	10.00	10.00
SST-2	$C = 20$ ($R_{\max} = 25$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	69.72 ± 29.19	$\mathbf{89.91 \pm 0.81}$	**14.0**	**39.4** MB	14.43	14.57	14.29
		FDA-RoLoRA E1TO20 ($k = 0.5$)	79.36 ± 16.38	89.39 ± 2.19	16.0 ± 2.8	45.0 MB	9.69	10.32	9.06
		RoLoRA E10	$\mathbf{89.62 \pm 1.87}$	89.62 ± 1.87	25.0	70.3 MB	10.00	10.00	10.00
	$C = 50$ ($R_{\max} = 20$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	$\mathbf{81.36 \pm 9.97}$	$\mathbf{85.95 \pm 5.60}$	**20.0**	**140.6** MB	14.57	14.60	14.55
		FDA-RoLoRA E1TO20 ($k = 0.5$)	66.46 ± 24.57	67.37 ± 23.27	**20.0**	**140.6** MB	8.70	8.40	9.00
		RoLoRA E10	53.33 ± 3.08	59.00 ± 11.11	**20.0**	**140.6** MB	10.00	10.00	10.00

Backup : Complete IID Benchmark Table (Table 5.1)

All Evaluated IID Configurations ($\alpha = 1000.0$) Across GLUE Benchmarks

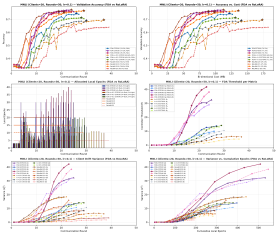
◂ Back to Results Summary ◂ Q&A

Task	Cohort	Methodology	Final Acc. (%)	Peak Acc. (%)	Rounds (R)	Total MB	Mean $\bar{e}$	Drift Var. ($\bar{\nu}$)
MNLI	$C = 20$ ($R_{\max} = 50$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	**76.86 ± 0.10**	**76.86 ± 0.10**	32.0 ± 4.2	150.0 MB	13.42	6.552
		FDA-RoLoRA E1TO20 ($k = 0.5$)	76.44 ± 0.18	76.45 ± 0.17	41.0 ± 4.2	192.2 MB	7.76	2.957
		RoLoRA E20	75.98 ± 0.58	76.28 ± 0.16	**22.0**	**103.1** MB	20.00	10.149
		RoLoRA E15	75.21 ± 2.67	75.34 ± 2.86	22.5 ± 2.1	105.5 MB	15.00	6.004
		RoLoRA E10	76.60 ± 0.85	76.60 ± 0.85	28.0 ± 1.4	131.2 MB	10.00	3.713
		RoLoRA E5	63.89 ± 14.80	68.52 ± 8.25	32.5 ± 9.2	152.3 MB	5.00	0.711
	$C = 50$ ($R_{\max} = 30$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	**69.85 ± 3.37**	**69.85 ± 3.37**	**30.0**	**351.6** MB	12.15	1.115
		FDA-RoLoRA E1TO20 ($k = 0.5$)	56.16 ± 7.99	56.16 ± 7.99	**30.0**	**351.6** MB	7.22	0.106
		RoLoRA E10	65.85 ± 2.85	65.85 ± 2.85	**30.0**	**351.6** MB	10.00	0.416
		RoLoRA E5	34.57 ± 2.43	35.81 ± 0.67	**30.0**	**351.6** MB	5.00	0.010
QNLI	$C = 20$ ($R_{\max} = 50$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	67.63 ± 22.28	**83.14 ± 0.34**	16.5 ± 3.5	77.3 MB	9.85	1.111
		FDA-RoLoRA E1TO20 ($k = 0.5$)	65.28 ± 22.37	80.61 ± 0.69	17.5 ± 0.7	82.0 MB	6.91	0.448
		RoLoRA E10	**75.03 ± 10.54**	81.83 ± 0.92	**10.0 ± 1.4**	**46.9** MB	10.00	0.977
		RoLoRA E5	71.72 ± 14.63	81.79 ± 0.39	16.5 ± 2.1	77.3 MB	5.00	0.373
	$C = 50$ ($R_{\max} = 30$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	82.47 ± 1.72	82.77 ± 2.14	**21.0 ± 5.7**	**246.1** MB	10.89	0.707
		FDA-RoLoRA E1TO20 ($k = 0.5$)	80.75 ± 0.17	80.96 ± 0.18	26.0 ± 4.2	304.7 MB	7.35	0.213
		RoLoRA E20	**87.94 ± 0.00**	**88.14 ± 0.00**	30.0	351.6 MB	20.00	11.343
		RoLoRA E10	82.00 ± 1.85	82.85 ± 1.61	24.5 ± 2.1	287.1 MB	10.00	0.435
		RoLoRA E5	78.82 ± 1.48	79.25 ± 0.87	30.0	351.6 MB	5.00	0.034
QQP	$C = 20$ ($R_{\max} = 50$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	**76.34 ± 5.33**	**78.57 ± 4.23**	17.5 ± 6.4	82.0 MB	10.69	1.576
		FDA-RoLoRA E1TO20 ($k = 0.5$)	72.96 ± 0.58	74.17 ± 1.12	14.0 ± 1.4	65.6 MB	6.50	0.543
		RoLoRA E10	75.47 ± 1.19	75.47 ± 1.19	**6.5 ± 0.7**	**30.5** MB	10.00	0.761
		RoLoRA E5	72.46 ± 1.17	73.58 ± 0.20	10.0	46.9 MB	5.00	0.323
	$C = 50$ ($R_{\max} = 30$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	72.65 ± 1.17	73.63 ± 0.03	**14.0**	**164.1** MB	9.14	0.220
		FDA-RoLoRA E1TO20 ($k = 0.5$)	72.08 ± 0.20	73.86 ± 0.39	18.0 ± 1.4	210.9 MB	6.25	0.206
		RoLoRA E10	**75.43 ± 1.07**	**78.79 ± 1.38**	30.0	351.6 MB	10.00	0.731
		RoLoRA E5	75.36 ± 0.50	75.36 ± 0.50	30.0	351.6 MB	5.00	0.221
SST-2	$C = 20$ ($R_{\max} = 50$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	91.80 ± 0.24	92.14 ± 0.24	20.5 ± 0.7	57.7 MB	11.11	0.654
		FDA-RoLoRA E1TO20 ($k = 0.5$)	87.61 ± 5.35	88.88 ± 3.73	17.5 ± 7.8	49.2 MB	5.82	0.113
		RoLoRA E20	**92.16 ± 0.13**	**92.55 ± 0.30**	**10.7 ± 1.5**	**30.0** MB	20.00	2.387
		RoLoRA E10	91.51 ± 0.32	91.80 ± 0.08	15.0 ± 1.4	42.2 MB	10.00	0.327
		RoLoRA E5	90.31 ± 1.22	90.60 ± 0.81	21.0 ± 1.4	59.1 MB	5.00	0.105
	$C = 50$ ($R_{\max} = 30$)	FDA-RoLoRA E1TO20 ($k = 1.0$)	89.68 ± 1.14	89.68 ± 1.14	20.0 ± 5.7	140.6 MB	10.78	0.063
		FDA-RoLoRA E1TO20 ($k = 0.5$)	86.81 ± 4.05	87.44 ± 3.16	25.0 ± 7.1	175.8 MB	6.91	0.019
		RoLoRA E20	**91.90 ± 0.37**	**92.01 ± 0.57**	**17.0 ± 11.4**	**119.5** MB	20.00	1.660
		RoLoRA E10	89.91 ± 0.49	90.54 ± 0.73	22.0	154.7 MB	10.00	0.109
		RoLoRA E5	87.33 ± 0.41	87.50 ± 0.65	26.0 ± 5.7	182.8 MB	5.00	0.013

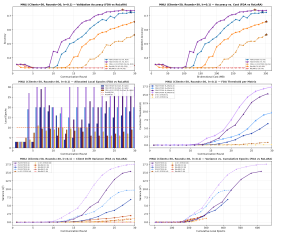
Backup : Additional IID Plots (MNLI & QNLI)

Multi-Panel Optimization Plots Under Homogeneous Consensus ($\alpha = 1000.0$)

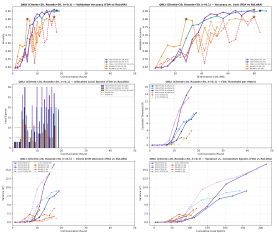
Back to IID Q&A



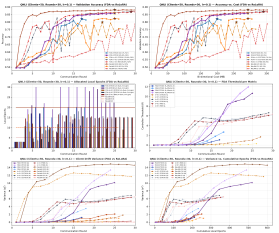
MNLI ($C = 20$, IID)



MNLI ($C = 50$, IID)



QNLI ($C = 20$, IID)

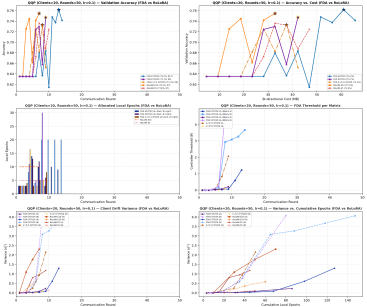


QNLI ($C = 50$, IID)

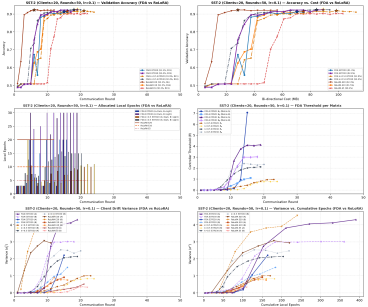
Backup : Additional IID Plots (QQP & SST-2)

Multi-Panel Optimization Plots Under Homogeneous Consensus ($\alpha = 1000.0$)

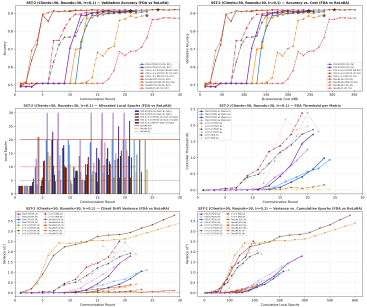
◂ Back to IID ◂ Q&A



QQP ($C = 20$, **IID**)



SST-2 ($C = 20$, **IID**)



SST-2 ($C = 50$, **IID**)